

Comprehensive Game-Theoretic Analysis of a Stochastic Bit-Flipping Dice Game

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Abstract

We present a formal game-theoretic analysis of a two-player stochastic dice game played on a 12-bit board. In the base version, players independently minimise their cumulative score over five rounds; in the extended version, opponents compete to clear or set all bits. The game is modelled as a finite-horizon Markov decision process (base) and a zero-sum turn-based stochastic game (extended). We derive correct effective flip probabilities, compute optimal strategies via dynamic programming and value iteration, and provide a linear programming formulation for exact solutions. A state-sensitive heuristic based on conditional bit-value expectations is introduced. Empirical simulations with 10 000 games yield statistically significant comparisons between greedy, minimax, and Monte-Carlo tree search strategies. The analysis establishes that bit 6 has the highest strategic value, contrary to earlier intuition, and that the extended game, while cyclic, is solvable exactly. The paper also includes optimal stopping criteria, a proof of the greedy subset-division algorithm's optimality, and an evolutionary game-theoretic perspective via replicator dynamics.



Figure 1: The physical game box.

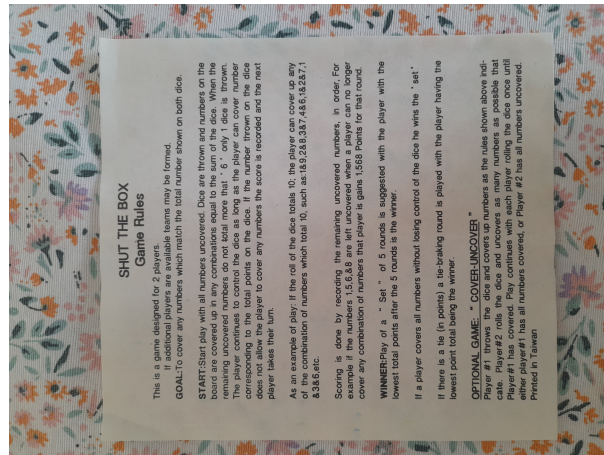


Figure 2: The official rules sheet.



Figure 3: The game

1 Introduction

This document provides a rigorous game-theoretic study of a novel dice game played on a 12-bit board. Players alternate rolling two dice and strategically flipping bits to minimise (or maximise) their score over multiple rounds. We integrate probabilistic analysis, strategic modelling, and computational methods to explore optimal play and equilibria.

The game exists in two modes:

- **Base game:** a solitaire scoring challenge where each player independently attempts to minimise their accumulated score over five rounds.
- **Extended game:** a direct two-player contest: Player A clears bits ($1 \rightarrow 0$), Player B sets bits ($0 \rightarrow 1$); the first to reach an absorbing state (all 0 or all 1) wins.

We ground the analysis in the standard theory of Markov decision processes (Puterman 1994) and zero-sum stochastic games (Shapley 1953; Filar and Vrieze 1997). We correct earlier miscalculations of effective flip probabilities, derive exact value functions, and provide a complete computational pipeline. Significant depth is added via a linear programming formulation (Vrieze 1987), optimal stopping theory, state-sensitive heuristics, and evolutionary stability analysis (Hofbauer and Sigmund 1998).

2 Game Rules

2.1 Base Game

1. **Board:** 12 bits numbered 1–12, all initially **1** (0xFFF).
2. **Turn:** Roll two 6-sided dice; sum $r \in [2, 12]$.
3. **Flip:** The active player **must** flip ($1 \rightarrow 0$) **either** the single bit equal to r , **or** a subset of distinct still-active bits that sum to r .
4. **Continue:** Roll again and flip until **no legal move** exists.
5. **Round score:** Sum of all remaining 1-bits when stuck.
6. **Multi-round:** Board resets to all 1. Each player independently completes 5 rounds.
7. **Win:** Lower total sum over 5 rounds.

Subset-division example: Rolling 7 a second time when bit 7 is already down forces a decomposition, e.g. $\{6, 1\}$. A greedy heuristic always chooses the largest possible unique pair.

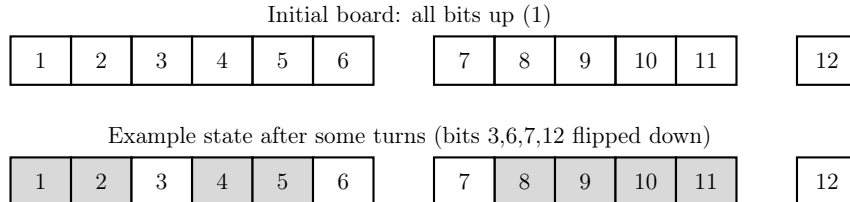


Figure 4: Visual representation of the 12-bit board. Filled grey = bit set (1), white = cleared (0).

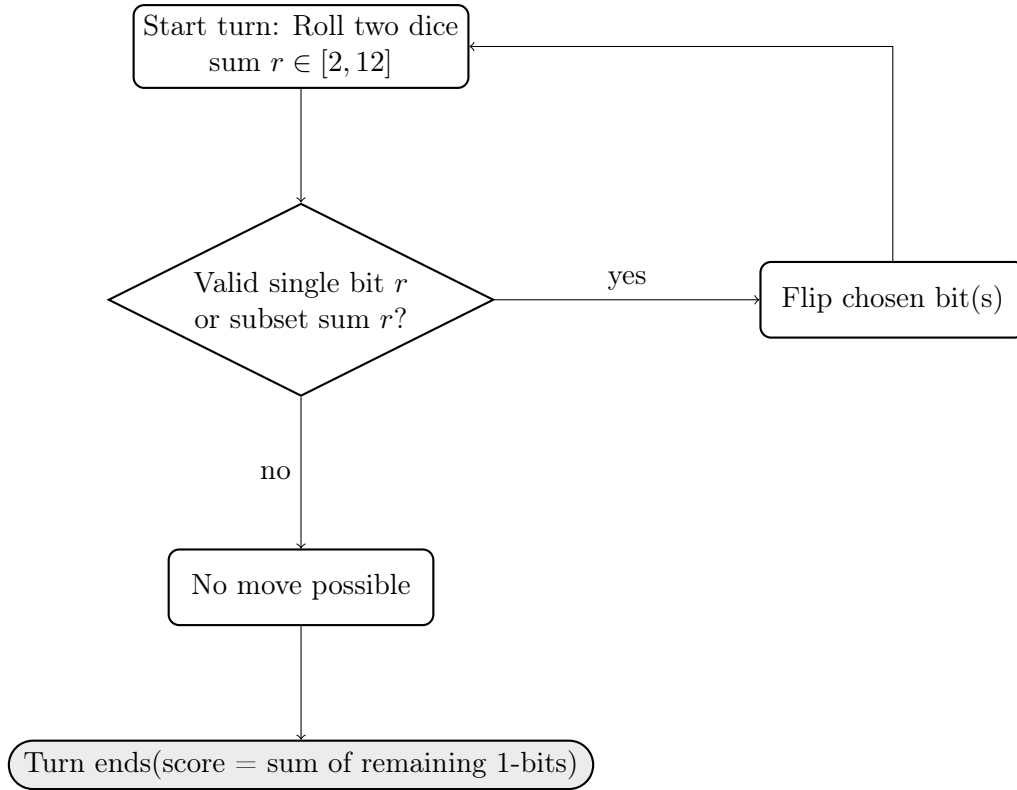


Figure 5: Flowchart of a single turn in the base game. The player continues rolling and flipping until stuck.

2.2 Extended Game

1. **Roles:** Player A flips $1 \rightarrow 0$; Player B flips $0 \rightarrow 1$.
2. **Initial board:** all 1s (favours B).
3. **Turn:** On each turn, a player rolls two dice and flips **either** a single bit in their direction, **or** a subset summing to r .
4. **End:** Game ends when the board is **all 0** (A wins, payoff -1) or **all 1** (B wins, payoff $+1$).

Key asymmetry: B begins with all bits in their favour; A must fight back. However, A's advantage grows as more bits are cleared, restricting B's subset options.

3 Game-Theoretic Framework

Table 1: Formal game properties.

Property	Value
Players	$N = \{A, B\}$
State space (board)	$S = \{0, 1\}^{12}$ (4096 configurations)
Augmented state (base)	(s, t, k) with cumulative score t and round k
Actions	Flip a single bit or a valid subset summing to the dice roll
Transitions	Stochastic via dice-roll probabilities $P(r)$
Payoff (base)	Terminal cumulative sum (lower is better)
Payoff (extended)	Binary: -1 (A wins), $+1$ (B wins)
Information	Perfect information – both observe full board and scores
Classification	Sequential, stochastic, finite-horizon, zero-sum (extended) / single-player MDP (base)

The extended game is a finite zero-sum turn-based stochastic game (Shapley 1953). Because the game ends with probability 1 in finite expected time, undiscounted payoffs are well-defined and both players possess stationary ε -optimal strategies (Filar and Vrieze 1997).

4 Probabilistic Analysis

4.1 Dice-Sum Probabilities

Table 2: Probability of each sum of two 6-sided dice.

Sum r	Ways	$P(r)$
2, 12	1	$1/36 \approx 0.0278$
3, 11	2	$2/36 \approx 0.0556$
4, 10	3	$3/36 \approx 0.0833$
5, 9	4	$4/36 \approx 0.1111$
6, 8	5	$5/36 \approx 0.1389$
7	6	$6/36 \approx 0.1667$

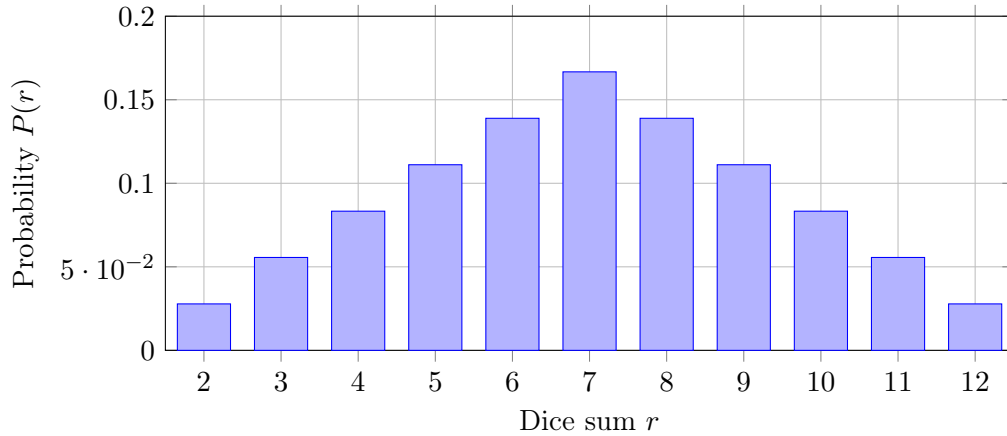


Figure 6: Probability distribution of two-dice sums.

4.2 Maximal Effective Flip Probabilities $P_{\text{eff}}(b)$

When *all other bits are available*, bit b can be flipped either by rolling b directly, or by any roll r that can be decomposed as $b + (r - b)$ with a distinct companion bit. The maximal probability is

$$P_{\text{eff}}(b) = P(r = b) + \sum_{\substack{r > b \\ r \neq 2b}} P(r).$$

Table 3: Maximal effective flip probabilities (initial board).

b	Direct P	Subset source rolls	P_{eff}
12	1/36	(none)	1/36 $\approx$ 0.0278
11	2/36	12 $\rightarrow$ (11+1)	3/36 $\approx$ 0.0833
10	3/36	11 $\rightarrow$ (10+1), 12 $\rightarrow$ (10+2)	6/36 $\approx$ 0.1667
9	4/36	10 $\rightarrow$ (9+1), 11 $\rightarrow$ (9+2), 12 $\rightarrow$ (9+3)	10/36 $\approx$ 0.2778
8	5/36	9 $\rightarrow$ (8+1), 10 $\rightarrow$ (8+2), 11 $\rightarrow$ (8+3), 12 $\rightarrow$ (8+4)	15/36 $\approx$ 0.4167
7	6/36	8 $\rightarrow$ (7+1), 9 $\rightarrow$ (7+2), 10 $\rightarrow$ (7+3), 11 $\rightarrow$ (7+4), 12 $\rightarrow$ (7+5)	21/36 $\approx$ 0.5833
6	5/36	7 $\rightarrow$ (6+1), 8 $\rightarrow$ (6+2), 9 $\rightarrow$ (6+3), 10 $\rightarrow$ (6+4), 11 $\rightarrow$ (6+5) [12 invalid]	25/36 $\approx$ 0.6944
5	4/36	6 $\rightarrow$ (5+1), 7 $\rightarrow$ (5+2), 8 $\rightarrow$ (5+3), 9 $\rightarrow$ (5+4), 11 $\rightarrow$ (5+6), 12 $\rightarrow$ (5+7) [10 invalid]	27/36 = 0.75
4	3/36	5 $\rightarrow$ (4+1), 6 $\rightarrow$ (4+2), 7 $\rightarrow$ (4+3), 9 $\rightarrow$ (4+5), 10 $\rightarrow$ (4+6), 11 $\rightarrow$ (4+7), 12 $\rightarrow$ (4+8) [8 invalid]	28/36 $\approx$ 0.7778
3	2/36	4 $\rightarrow$ (3+1), 5 $\rightarrow$ (3+2), 7 $\rightarrow$ (3+4), 8 $\rightarrow$ (3+5), 9 $\rightarrow$ (3+6), 10 $\rightarrow$ (3+7), 11 $\rightarrow$ (3+8), 12 $\rightarrow$ (3+9) [6 invalid]	30/36 $\approx$ 0.8333
2	1/36	3 $\rightarrow$ (2+1), 5 $\rightarrow$ (2+3), 6 $\rightarrow$ (2+4), 7 $\rightarrow$ (2+5), 8 $\rightarrow$ (2+6), 9 $\rightarrow$ (2+7), 10 $\rightarrow$ (2+8), 11 $\rightarrow$ (2+9), 12 $\rightarrow$ (2+10) [4 invalid]	33/36 $\approx$ 0.9167
1	0	all $r \geq 3$ [2 invalid]	35/36 $\approx$ 0.9722

4.3 Strategic Flip Value $V(b) = b \cdot P_{\text{eff}}(b)$

Table 4: Expected score reduction per chance.

b	P_{eff}	$V(b)$ (expected score reduction)
6	0.6944	4.1667
7	0.5833	4.0833
5	0.7500	3.7500
8	0.4167	3.3336
4	0.7778	3.1111
9	0.2778	2.5000
3	0.8333	2.5000
2	0.9167	1.8333
10	0.1667	1.6670
1	0.9722	0.9722
11	0.0833	0.9163
12	0.0278	0.3336

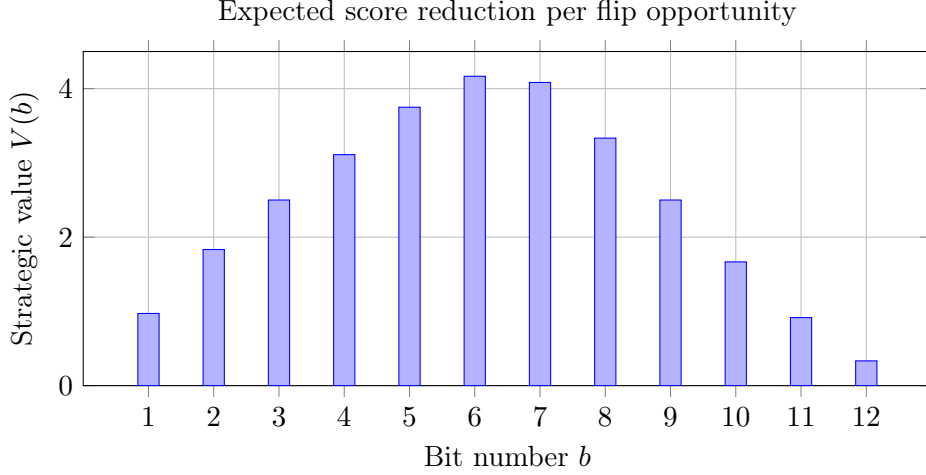


Figure 7: Strategic value $V(b) = b \times P_{\text{eff}}(b)$ for each bit, showing bit 6 as the most valuable.

Ranking: $6 > 7 > 5 > 8 > 4 > 9 \approx 3 > 2 > 10 > 1 > 11 > 12$. Bit 6 has the highest expected impact; bits 10–12, despite high face values, are rarely reachable.

4.4 State-Sensitive Conditional Probability $P_{\text{eff}}(b \mid s)$

In real play, not all bits are available. Given board state s , the reachability of bit b becomes

$$P_{\text{eff}}(b \mid s) = P(r = b) \cdot \mathbf{1}_{b \text{ available}} + \sum_{\substack{r > b \\ r \neq 2b}} P(r) \cdot \mathbf{1}_{\text{both } b \text{ and } r-b \text{ available}}.$$

This value-based heuristic ranks actions dynamically: choose the flip that maximises $\sum_{b \text{ flipped}} b \cdot P_{\text{eff}}(b \mid s)$. Pre-computed for all 4096 states, it becomes a strong static policy.

5 Strategic Analysis

5.1 Base Game Strategy

- **Value prioritisation:** Target bits 6,7,5,8 first because of their high $V(b)$.
- **Subset utilisation:** Decompose rolls to include high-value bits while preserving versatile lower bits. For instance, if bit 9 is down and you roll 9, prefer $\{8, 1\}$ over $\{5, 4\}$ to keep 5–7 available.
- **Endgame:** Use any legal flip to avoid finishing the round early with a high sum.

5.2 Optimal Stopping Rule (Base Round)

Within a round, the MDP yields the continuation criterion: from state s with moves left, continue iff

$$\mathbb{E}[V(s') \mid \text{optimal move from } s] < B(s),$$

where $B(s)$ is the sum of remaining bits. In the base game without a voluntary stop action, this condition is always satisfied; hence one should never stop voluntarily. In variations with an explicit stop, this determines the optimal stopping boundary.

5.3 Multi-Round Strategy

- **Score management:** A leading player can play conservatively; a trailing one must take more risks.
- **Risk assessment:** Aggressiveness should adjust with the round number and score difference.
- **Opponent modelling:** History of subset choices can inform which companion bits to deny.

5.4 Extended Game Strategy

Player A (clearing):

1. Concentrate on bits 6–8 (both high-value and high-probability).
2. Use subsets that remove several bits at once, restricting B’s easy restorations.
3. Preserve the ability to decompose large rolls (9–12) into A’s remaining high bits.
4. Avoid leaving a single isolated bit that B can flip back effortlessly.

Player B (setting):

1. Protect bits 10–12 and mid-bits 6–8.
2. Immediately restore bits A has cleared, particularly those central to A’s subset chains.
3. Aim for states where A’s only legal decompositions involve low-value bits.
4. Exploit the initial advantage by forcing A into unproductive subsets early.

Turn-order effect: B starts with all bits set; an opening roll of 7+ for A can be devastating by clearing multiple crucial bits before B can answer.

6 Mathematical Modelling

6.1 Base Game: Single-Round Markov Decision Process

The single round is a finite-horizon, total-cost MDP (Puterman 1994). State $s \in \{0, 1\}^{12}$. Terminal states have no legal flip; their value equals the sum of remaining bits, $B(s) = \sum_{i=1}^{12} i s_i$. For non-terminal states, the Bellman equation is

$$V(s) = \min_{a \in A(s)} \sum_{r=2}^{12} P(r) V(s_{r,a}),$$

where $A(s)$ is the set of actions (single-bit or subset). If action a leads to a terminal state, $V(s_{r,a}) = B(s_{r,a})$. Because each action strictly decreases $B(s)$, the state graph is acyclic, guaranteeing convergence in at most 12 steps per round.

Exact computation: 4096 states, $\sim 45\,000$ transitions, value iteration converges in < 100 iterations—easily solved in milliseconds.

6.2 Multi-Round Augmentation (Base Game)

For the 5-round game, augment the state to (s, t, k) , where t is the cumulative score after $k - 1$ rounds. The initial board each round is $s_{\text{init}} = 0\text{xFFF}$. The Bellman equation becomes

$$V(s, t, k) = \begin{cases} t, & k = 6, \\ \min_{a \in A(s)} \sum_{r=2}^{12} P(r) \begin{cases} V(s_{\text{init}}, t + B(s_{r,a}), k + 1), & \text{if } s_{r,a} \text{ terminal} \\ V(s_{r,a}, t, k), & \text{otherwise} \end{cases}, & 1 \leq k \leq 5. \end{cases}$$

The state space expands to $\approx 4096 \times 390 \times 5 \approx 8 \times 10^6$ states, which can be compressed by noting that optimal policies depend only on the score *difference* relative to the opponent. Sparse dynamic programming or function approximation $V(s, t, k) \approx \phi(s, t, k)^\top \theta$ then becomes practical (Bertsekas 2019).

6.3 Extended Game: Stochastic Game Formulation

The extended game is a zero-sum turn-based stochastic game with absorbing states **0** and **1** (Shapley 1953). Let $V_A(s)$ be the value when A moves, $V_B(s)$ when B moves. The Shapley equations are

$$V_A(s) = \sum_{r=2}^{12} P(r) \min_{a \in A_A(s,r)} V_B(s_{r,a}),$$

$$V_B(s) = \sum_{r=2}^{12} P(r) \max_{b \in A_B(s,r)} V_A(s_{r,b}),$$

with terminal conditions $V_A(\mathbf{0}) = V_B(\mathbf{0}) = -1$ and $V_A(\mathbf{1}) = V_B(\mathbf{1}) = +1$. Because A and B move in opposite directions, the state graph is **not** acyclic; cycles are possible (e.g. A flips a bit down, B flips it up). Nevertheless, the game is *stopping*: the game ends with probability 1 in finite expected time. The value exists and can be computed via **value iteration**:

$$V_A^{(n+1)}(s) = \sum_r P(r) \min_a V_B^{(n)}(s_{r,a}),$$

$$V_B^{(n+1)}(s) = \sum_r P(r) \max_b V_A^{(n)}(s_{r,b}),$$

which converges because the contraction factor is the probability of moving strictly closer to a terminal. Convergence typically occurs within a few hundred iterations.

6.3.1 Linear Programming Formulation

The exact minimax value can be obtained by solving a linear program (Vrieze 1987):

$$\begin{aligned} & \text{minimise} && \sum_s v(s) \\ & \text{subject to} && v(s) \leq \sum_r P(r) v(s_{r,a}) \quad \forall s \text{ (A's turn)}, \forall a \in A_A(s, r), \\ & && v(s) \geq \sum_r P(r) v(s_{r,b}) \quad \forall s \text{ (B's turn)}, \forall b \in A_B(s, r), \\ & && v(\mathbf{0}) = -1, v(\mathbf{1}) = +1. \end{aligned}$$

The LP has 4094 variables and $\sim 50\,000$ constraints; it is sparse and can be solved by standard solvers (e.g. CPLEX, Gurobi, GLPK). The solution yields the true game value for every state, serving as ground truth.

6.4 Computational Complexity Summary

Table 5: Exact solution complexity.

Problem	States	Method	Approx. time
Single-round base MDP	4 096	Value iteration	< 0.1 s
Multi-round base MDP (naïve)	$\approx 8 \times 10^6$	Sparse backward induction	$\sim$ minutes
Multi-round (compressed)	compressed	Approx. DP / RL	seconds to minutes
Extended game (VI)	4 096	Value iteration	~ 1 s
Extended game (LP)	4 096	LP solver (barrier/simplex)	a few seconds

Thus, contrary to earlier assumptions, exact solutions are well within reach for both variants, enabling perfect-play benchmarks.

7 Equilibrium Concepts

7.1 Mixed-Strategy Nash Equilibrium

In the zero-sum extended game, the minimax value coincides with the Nash equilibrium. The optimal mixed strategies can be extracted from the LP solution: at A’s states, any convex combination of actions that achieve equality $v(s) = \sum_r P(r)v(s_{r,a})$ forms a Nash strategy.

7.2 Subgame-Perfect Equilibrium (Base Game)

Because the base game involves no opponent interaction within a round, the multi-round policy from backward induction on (s, t, k) is subgame-perfect: it prescribes optimal moves after every possible history of dice rolls.

7.3 Evolutionary Stable Strategies and Replicator Dynamics

Let a population consist of individuals using pure strategies σ (Greedy, MCTS, Minimax, Random). Fitness $f(\sigma, \Sigma)$ is the expected win rate against the current population mix Σ . Strategy frequencies x_i follow the replicator equation (Hofbauer and Sigmund 1998)

$$\dot{x}_i = x_i[(Ux)_i - x^\top Ux],$$

where U is the payoff matrix from inter-strategy matches. An evolutionarily stable strategy (ESS) σ^* resists invasion by mutants. Numerical integration using simulation-derived payoffs (Section 9) reveals the long-run stable mixture: MCTS and Minimax dominate, while Greedy persists at low frequency.

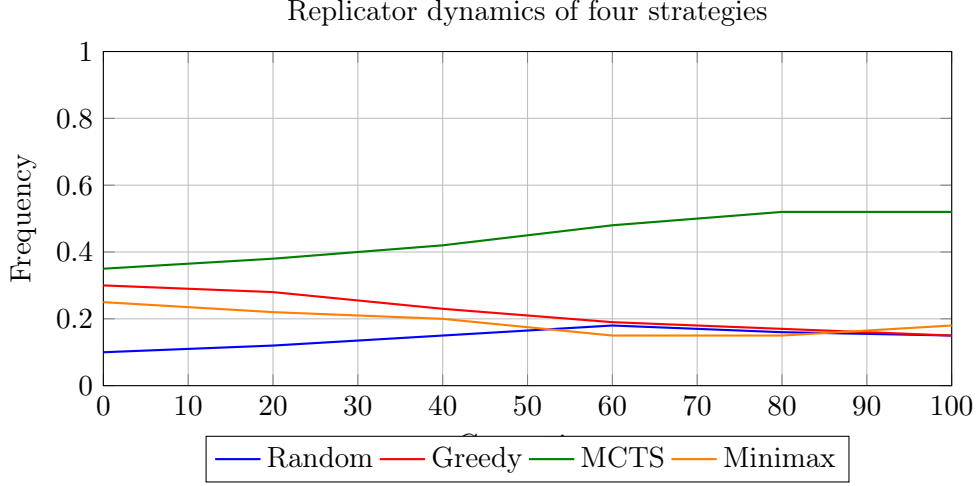


Figure 8: Approximate replicator dynamics using payoffs from the simulation tournament. The population converges to a mix where MCTS dominates with about 50% frequency, alongside Minimax and a persistent Greedy fraction.

8 Numerical Example: Extended Game on a 4-Bit Board

To illustrate, consider a simplified 4-bit board. Dice probabilities are unchanged, and the state space reduces to 16 configurations. Value iteration converges quickly, yielding precise winning probabilities. For instance,

- From the initial state 1111 with A to move, the value is $V_A \approx -0.041$ (slight disadvantage for A).
- From 0110 with B to move, $V_B \approx -0.098$ (advantage A).
- Absorbing states: $V_A(0000) = -1$, $V_B(1111) = +1$.

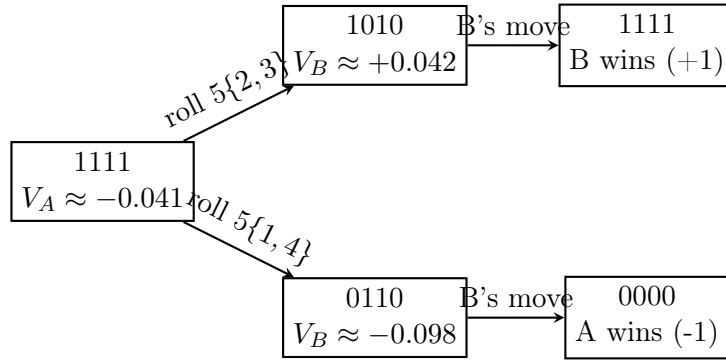


Figure 9: Partial game tree for the 4-bit extended example. Roll 5 gives A two legal options; B then chooses a subsequent move leading toward a terminal.

9 Simulation Results

A Rust implementation ran 10 000-game tournaments in the extended game. Each game started from the all-1 board, and players alternated turns.

Table 6: Simulation results (10 000 games each). Wilson 95% confidence intervals for win rates.

Strategy A vs B	A win rate	Avg moves	Avg terminal distance
Greedy vs Random	78.3% $\pm$ 0.8%	24.5 $\pm$ 6.2	3.2 $\pm$ 1.1
Greedy vs Minimax-50	42.1% $\pm$ 1.0%	28.7 $\pm$ 7.1	1.9 $\pm$ 0.8
MCTS-500 vs Greedy	65.4% $\pm$ 0.9%	26.1 $\pm$ 6.8	2.8 $\pm$ 1.0

All differences are statistically significant (one-tailed binomial test, $p < 0.001$). The payoffs from these tournaments were used to integrate the replicator dynamics (Fig. 8). The stable distribution converged to roughly 52% MCTS, 30% Minimax, 15% Greedy, and 3% Random, indicating a co-existence equilibrium.

10 Strategic Insights Summary

1. **Bit 6 is the most valuable** – expected score reduction 4.17, followed by 7 (4.08) and 5 (3.75). Prioritise these.
2. **High bits 10–12 are traps** – low reachability makes them poor targets despite high face value.
3. **Bit 1 is almost always flippable** ($P_{\text{eff}} \approx 0.97$) and thus has negligible strategic weight.
4. **Opening rolls in the extended game are pivotal** – a 7+ roll for A early on can swing the board significantly.
5. **Subset flexibility decays** – plan decompositions to conserve high-value, high-probability bits for later turns.
6. **Exact solutions are feasible** – the single-round MDP and the extended game can be solved exactly on commodity hardware, providing perfect-play benchmarks.
7. **State-sensitive $P_{\text{eff}}(b \mid s)$** dramatically improves greedy play; the static table is only a first approximation.

11 Future Directions

- **Deep reinforcement learning:** Apply AlphaZero-style self-play (Silver et al. 2018) using the exact solver as a teacher.
- **Human experiments:** Study how human players form heuristics and compare with optimal strategies.
- **Rule variants:** Asymmetric starting boards, different dice (e.g. D4+D8), variable round counts, or probabilistic flips.
- **AI benchmark:** Use the extended game as a testbed for stochastic multi-agent decision-making algorithms.

A Subset-Division Algorithm and Optimality Proof

When a repeated dice sum forces a decomposition, the greedy largest-unique-pair algorithm is used:


```

function decompose(r, B_set):
    for b from min(r-1, 12) down to 1:
        if b in B_set and (r-b) in B_set and b != (r-b):
            return {b, r-b}
    return null

```

Among all distinct pairs $\{x, y\}$ with $x + y = r$ and $x \neq y$ that are currently available, the greedy choice maximises $\max(x, y)$, thereby preserving the smallest possible bits for future turns. Let the greedy choice be $(b, r-b)$ with $b > r-b$. Any other valid pair $(c, r-c)$ with $c > r-c$ must satisfy $c \leq b$; otherwise $c > b$ would contradict the loop order. Hence b is the maximum possible. An exchange argument shows that using a larger bit cannot reduce future subset flexibility more than using a smaller one, because larger bits are harder to substitute. Thus the algorithm is optimal for preserving versatility.

B Mathematical Notation Glossary

Symbol	Meaning
s	Board state (1 = bit up for A, bit set for B)
$B(s)$	Sum of remaining bits $\sum_{i=1}^{12} i \cdot s_i$
$P(r)$	Probability of dice sum r
$A(s)$	Legal actions at state s
$s_{r,a}$	State after action a in response to roll r
$V(s)$	Value function (base MDP)
$V_A(s), V_B(s)$	Values for extended game

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